Week 6 progressive hints

Use the hints in order. After each hint, return to your own work and continue until you reach a new, specific impasse.

W06-Q01-PA

1. Write the column gradient and solve its two simultaneous equations.
2. The Hessian is constant. Use its leading principal minors, eigenvalues or a completed-square argument to determine definiteness.

W06-Q01-PB

1. The first-order condition is necessary for an interior differentiable local optimum, but it does not say which kind of candidate has been found.
2. Ask whether the Hessian classification holds only at the candidate or on all of the convex domain $\mathbb{R}^2$.

W06-Q02-PA

1. Set both entries of the column gradient equal to zero.
2. Solve the resulting 2×2 linear system, then classify the constant symmetric Hessian by its leading principal minors.

W06-Q02-PB

1. A negative definite Hessian everywhere on a convex domain establishes strict concavity, not merely local curvature at one point.
2. Combine strict concavity with the stationary point and then substitute the point into g to obtain the objective value.

W06-Q03-PA

1. The stationary equations separate: one involves x^2, and the other sets y to zero.
2. Evaluate the diagonal Hessian at both candidates; opposite signs imply a saddle.
3. To decide global existence, hold $y = 0$ and consider what happens as x tends to large positive and negative values.

W06-Q04-PA

1. Compute the gradient and Hessian before substituting the stationary point.
2. A zero Hessian is neither positive nor negative definite. Return to the definition of the objective and use the signs of fourth powers.
3. Check when both fourth powers can sum to zero, and examine behaviour as either coordinate grows without bound.

W06-Q05-PA

1. The Hessian is twice the symmetric matrix of the quadratic form. Its eigenvalues can be read from the directions $[1, 1]^\mathsf{T}$ and $[1, -1]^\mathsf{T}$.
2. Separate the cases in which both eigenvalues are negative, one is zero, or they have opposite signs.
3. At the boundary values of a, rewrite the objective as the negative of one square. Outside the boundary, test the two eigenvector directions.

W06-Q06-PA

1. Compute $\bar{x}$, then list the four deviations from it.
2. Square and sum the deviations, and divide by $n - 1 = 3$, not by n.

W06-Q06-PB

1. For a normal i.i.d. sample, $(n-1)S^2/\sigma^2$ has a chi-squared distribution with $n-1$ degrees of freedom.
2. Rewrite $S^2 \leq c$ as an inequality for $9S^2/4$.
3. Match the resulting threshold to the supplied 0.95 quantile and solve the equality for c.

W06-Q07-PA

1. A sum of independent normal random variables is normal. Find the mean and variance of both $\bar{X}$ and T.
2. The standard deviations are different: use $\sigma/\sqrt{n}$ for $\bar{X}$ and $\sqrt{n}\,\sigma$ for T.
3. Standardise each boundary, use symmetry of Φ, and keep the two requested probabilities separate.

W06-Q07-PB

1. A Bernoulli(0.5) variable has mean 0.5 and variance 0.25.
2. The standard deviation of $\bar{Y}$ is $\sqrt{0.25/100}$. Standardise both endpoints.
3. Apply the large-sample normal approximation to the standardised sample mean and use symmetry around zero.

W06-Q08-PA

1. For the stated event, substitute $\sigma^2 = 25$ and $\varepsilon = 1$ into the supplied bound.
2. Solve $25/n \leq 0.04$, remembering that n must be a positive integer.
3. The bound establishes concentration around the mean; it contains no claim about a normal shape.

W06-Q08-PB

1. The law of large numbers concerns convergence in probability of $\bar{X}$ to μ.
2. The central limit theorem concerns $\sqrt{n}(\bar{X} - \mu)/\sigma$, not the unscaled equality $\bar{X} = \mu$.
3. Separate a claim about the sampling distribution of the mean from a claim about the distribution of each observation.

W06-Q09-PA

1. Evaluate F at the stated point. Then compute F_x and F_θ, stacking them in the coordinate order (x, θ).
2. To define $x = x(\theta)$ locally, check the derivative with respect to the variable being solved for and recall that a polynomial is continuously differentiable.
3. Along $F(x, \theta) = 0$, write $\mathrm{d}F = F_x\,\mathrm{d}x + F_\theta\,\mathrm{d}\theta = 0$, then isolate $\mathrm{d}x/\mathrm{d}\theta$ before evaluating.

W06-Q10-PA

1. In the symmetric matrix, the cross term is $2a_{12}uv$.
2. For a symmetric 2×2 matrix, positive leading principal minors establish positive definiteness.
3. Relate positive definiteness of the constant Hessian to strict convexity on the convex domain.

W06-Q11-PA

1. With exponential rate 2, first write $\mathbb{E}[X]$ and $\mathrm{Var}(X)$. Independence makes the conditional moments of X given B equal to its unconditional moments.

2. When conditioning on B, treat B as known. Then apply iterated expectation to the resulting random variable $\mathbb{E}[Y \mid B]$.
3. Use both terms in $\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid B)] + \text{Var}(\mathbb{E}[Y \mid B])$, remembering that scaling by 3 scales variance by 3^2.